

AI Development Log

[Project Github repository](#)

[Project demo](#)

Tooling

The project was developed with Claude Code as the primary AI coding assistant.

AI was used for planning the application architecture, implementing backend and frontend features, generating and updating documentation, writing tests, debugging and reviewing seed data.

Claude Sonnet 5 was used for semantic search.

Data Strategy

The initial hardware dataset came from seed.json. An important part of the development process was auditing the dataset before treating it as correct input.

The model identified a few data quality problems:

- duplicate IDs
- inconsistent casing
- misspelled brand name
- inconsistent date formatting
- a future purchase date
- missing or invalid hardware information

Instead of ignoring or inventing missing values the data was split into:

- issues that could be corrected with high confidence
- issues where the intended value couldn't be determined (all of them were manually reviewed before dropping)

For example:

- 'Appel' was corrected to 'Apple'
- '22-05-2023' was converted to '2023-05-22'
- the future '2027-10-10' purchase date was not changed to a guessed date
- a duplicate ID was not reassigned and the row was dropped
- an Unknown Device row was dropped

This resulted in 8 clean rows from the original 11. All dropped rows were manually reviewed before dropping. If the dataset would be larger I would ask AI to group the rows by issue (e.g. missing ID: 31 rows, future date: 132 rows, duplicate: 513 rows) and then I would decide what to do with each group.

The AI analysis was supported by manual overview of the data. The model didn't identify a second data-integration issue: 'notes' and 'history' represented the same kind of free-text device context. I asked the model to merge both fields into a single 'notes' column.

Prompt Trail

Full record of the prompts used to build this app can be found [here](#).

The "Correction"

The clearest example happened during the semantic-search development:

Initially Claude received the following fields: : *id*, *name*, *brand*, *notes* and *status*. The *purchase_date* field was missing. The feature initially appeared to work for queries such as "I need something to test my mobile app on", but a query such as: "devices that had been purchased before 2023" returned no results even though matching records existed. The failure did not make sense from the user's perspective because the database contained matching purchase dates. I traced the data being passed to the LLM and found that *purchase_date* was never included in the model's input. The problem wasn't the natural-language prompt or the model's reasoning, it was an incomplete data migration into the LLM context. The implementation was changed so the semantic-search payload includes: *id*, *name*, *brand*, *purchase_date*, *notes* and *status*. The system prompt was also updated to explicitly tell Claude that *purchase-date* and age-based queries are supported. A regression test was added to inspect the actual payload sent to the mocked Claude client and assert that *purchase_date*, *brand*, *notes* and *status* are present.

There were also some smaller examples where I had to review and correct the AI-generated changes:

- *history* field in the seed data: Claude initially overlooked this field during the data migration. I noticed that the information would be lost and asked Claude to unify *history* with *notes*, so both fields would be preserved in the final notes column
- Python virtual environment: Claude Code created the virtual environment but initially did not add it to .gitignore. I caught this during my repository review and asked it to add it to .gitignore to prevent the generated environment from being tracked